

Associate Professor, Beihang University, China

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RESEARCH INTEREST

- **Intelligent Software Engineering:** automated program repair, software testing.
- **Software Security:** vulnerability detection and fix, software supply chain security
- **AI Engineering:** robust AI, AI security

EDUCATION

- **National University of Singapore, School of Computing** Singapore
Ph.D. *Fall 2016 – Fall 2021*
Advisor: Abhik Roychoudhury
- **Shandong University, Computer Science** Shandong, China
Bachelor (Elite class) *Fall, 2012 – Jun, 2016*
Advisor: Lei Ju

WORK EXPERIENCE

- **Associate Professor— Beihang University** 2022, 01 – now
School of Software
- **Postdoctor Researcher— National University of Singapore** 2021, 04 – 2021, 12
Conduct research on program transformation.
- **Research Intern — Microsoft Research** 2020, 01 – 2020, 06
Program synthesis to automatically generate edit suggestions in Visual Studio.
- **Research Intern — Fujitsu Laboratories of American** 2018, 09 – 2018, 12
Enhance the robustness of AI models via data augmentation.
- **System Engineer Intern — Alibaba** 2015, 07 – 2015, 10
Use Security Enhanced Android (SEAndroid) to increase Android security.

RESEARCH PROJECTS

- **Deep learning for bug detection and repair:** systematically designed a set of techniques, using deep learning, large language model, test generation, symbolic reasoning and semi-supervised approaches, to automatically detect and fix program bugs and vulnerabilities. We showed that our approach can fix the bugs detected by OSS-Fuzz. The designed tool has been integrated into VS IntelliCode.
- **Scalable code checking:** design code checking platform via DSL-based code property graph analysis. We showed that our approach can efficiently detect API misuse, PyTorch bugs, vulnerabilities, concurrency bugs and etc.
- **AI model engineering:** design techniques to enhance the security and explainability of AI models. Won ACT SIGSOFT distinguished paper award at ICSE 2024.

REPRESENTATIVE PUBLICATIONS

- **Modularizing while Training: A New Paradigm for Modularizing DNN Models** ICSE'24
Binhang Qi, Hailong Sun, Hongyu Zhang, Ruobing Zhao, Xiang Gao[#] (CCF-A)
International Conference on Software Engineering, 2024
ACM SIGSOFT Distinguished Paper Award
- **Investigating and Detecting Silent Bugs in PyTorch Programs** SANER'24
Shuo Hong, Hailong Sun, Xiang Gao[#], Shin Hwei Tan (CCF-A)
International Conference on Software Analysis, Evolution and Reengineering, 2024
IEEE TCSE Distinguished Paper Award
- **Automated Fixing of Web UI Tests via Iterative Element Matching** ASE'23
Yuanzhang Lin, Guoyao Wen, Xiang Gao[#] (CCF-A)
International Conference on Automated Software Engineering, 2023
- **Automated Repair of Programs from Large Language Models** ICSE'23
Zhiyu Fan, Xiang Gao[#], Martin Mirchev, Abhik Roychoudhury, Shin Hwei Tan (CCF-A)
International Conference on Software Engineering (ICSE) 2023
- **Patching Weak CNN Models through Modularization and Composition** ASE'22
Binhang Qi, Hailong Sun[#], Xiang Gao[#], Hongyu Zhang (CCF-A)
International Conference on Automated Software Engineering, 2022
- **Program Vulnerability Repair via Inductive Inference** ISSTA'22
Yuntong Zhang, Xiang Gao[#], Gregory J. Duck, Abhik Roychoudhury (CCF-A)
International Symposium on Software Testing and Analysis, 2022
- **Automated Patch Backporting in Linux (Experience Paper)** ISSTA'21
R. Shariffdeen, X. Gao*, G. J. Duck, S. Tan, J. Lawall, A. Roychoudhury (*co-first author)* (CCF-A)
International Symposium on Software Testing and Analysis, 2021
ACM SIGSOFT Distinguished Artifact Award
- **Beyond Tests: Program Vulnerability Repair via Crash Constraint Extraction** TOSEM'21
Xiang Gao, Bo Wang, Gregory J. Duck, Ruyi Ji, Yingfei Xiong, Abhik Roychoudhury (CCF-A)
Transactions on Software Engineering and Methodology, 2020
- **Feedback-Driven Semi-Supervised Synthesis of Program Transformations** OOPSLA'20
X. Gao, S. Barke, A. Radhakrishna, G. Soares, S. Gulwani, A. Leung, N. Nagappan, A. Tiwari (CCF-A)
Object-Oriented Programming, Systems, Languages, and Applications, 2020
- **Binary Rewriting without Control Flow Recovery** PLDI'20
Gregory J. Duck, Xiang Gao, Abhik Roychoudhury (CCF-A)
Programming Language Design and Implementation, 2020.
- **Fuzz Testing based Data Augmentation to Improve Robustness of Deep Neural Networks** ICSE'20
Xiang Gao, Ripon K. Saha, Mukul R. Prasad, Abhik Roychoudhury (CCF-A)
International Conference on Software Engineering, 2020.
- **Crash-avoiding Program Repair** ISSTA'19
Xiang Gao, Sergey Mechtaev, Abhik Roychoudhury (CCF-A)
International Symposium on Software Testing and Analysis, 2019.
- **Android Testing via Synthetic Symbolic Execution** ASE'18
Xiang Gao, Shin Hwei Tan, Zhen Dong, Abhik Roychoudhury (CCF-A)
International Conference on Automated Software Engineering, 2018.

TEACHING

• Intelligent Software Engineering Artificial intelligence for software engineering.	Beihang University 2024, Spring
• Open-Source Software Development Understanding, utilizing, participating, outlook open-source	Beihang University 2022-2023, Fall
• Object-Oriented Programming Object-oriented programming ideas, methods and technologies	Beihang University 2022-2023, Fall
• Software Testing, Formal Methods, Computer Organization Teaching assistant, design courses project, Lab tutor.	National University of Singapore 2017– 2019

SELECTED GRANT

• National Natural Science Foundation of China, 300K Software supply chain vulnerability detection and repair	2023-2025
• Ministry of Industry and Information Technology QM program, 4M Software, system and AI security	2023-2025
• Beihang-Huawei joint laboratory, 800K API misuse detection via code property graph query	2023-2024
• Beihang-Huawei joint laboratory, 800K Security risk warning method of open source software supply chain	2023-2024
• Ministry of Industry and Information, 1.8M Open source software culture training and promotion	2022-2024

SELECTED AWARDS

• ACM SIGSOFT Distinguished Paper Award at ICSE	2024
• IEEE TCSE Distinguished Paper Award at SANER	2024
• Excellent Faculty Award, School of Software, Beihang University	2023
• ACM SIGSOFT Distinguished Artifact Award at ISSTA	2021
• Dean's Graduate Research Excellence Award, NUS	2019
• Research Achievement Award, NUS	2018
• President's Graduate Fellowship, Singapore	2016 – 2020